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Next-Gen Speech Disorder Diagnostics: CNN Methods for Dysarthria Classification

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Abstract— Neurological disorders such as amyotrophic lateral sclerosis (ALS) and cerebral palsy (CP) are common causes of dysarthria, the subject of this research article. Dysarthria has a significant impact on communication capacity, making early diagnosis essential for effective therapies and improved patient outcomes. The research used a Convolutional Neural Network (CNN) to categorise speech using the TORGO database, which includes 2000 samples of individuals with and without dysarthria, representing different genders. There are 500 samples from each gender in the collection, representing dysarthric and non-dysarthric people who were recorded during different sessions. Highlighting the promise of deep learning technology in aiding early identification of dysarthria and enabling rapid therapy for affected individuals, the CNN-based approach exhibits an amazing 96% accuracy in discriminating between dysarthric and non-dysarthric cases. This study offers a promising chance to enhance the quality of life for those with speech impairments while also making substantial contributions to the development of diagnostic capabilities.

Keywords— Artificial Intelligence, Deep Learning, Convolutional Neural Network (CNN) model, Model Training, Dysarthria Classification

I. INTRODUCTION

People with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS) commonly struggle with dysarthria, which makes it difficult for them to communicate effectively. In order to improve patient outcomes, this study emphasises the critical need of early dysarthria diagnosis in allowing for timely treatment. This study makes use of the TORGO database, a large-scale collection of 2000 samples of both dysarthria and non-dysarthria individuals. The samples are recorded over the course of many sessions and then sorted according to gender. By classifying speech using a Convolutional Neural Network (CNN), the proposed method exemplifies how deep learning technology may distinguish between non-dysarthric and dysarthric scenarios. With 500 samples of each gender, the dataset provides a diverse and representative sample of people with and without dysarthria. The CNN-based method significantly advances the field of dysarthria diagnosis, with an impressive success rate of 96% in accurately detecting the condition. According to the research on digital speech biomarkers by Kovac et al., understanding hypokinetic dysarthria patterns in a wide variety of languages requires advancements in dysarthria detection. SFurthermore, Wang et al. stress the need of enhancing pre-trained ASR systems to identify dysarthric speech. Barlow et al.'s assessment of deep learning software for speech issue detection shares this opinion. Following the lead of Rao and Meher [3] and contributing to the ongoing discussion on data-driven vs. model-driven approaches in cognitive speech processing [4], the proposed study fits within the broader framework of automated diagnostic models for various disorders. The versatility of speech analysis in medical diagnostics is further demonstrated by Zhang et al., who provide an embedding-based approach to detect Wilson's disease from unstructured linked speech. Veetil et al.'s research on language-independent voice data-driven detection of Parkinson's disease and Pragadeeswaran and Kannimuthu's Adaptive Intelligent Polar Bear Optimization-Quantized Contempo Neural Network model for Parkinson's disease diagnosis both demonstrate the promise of using AI to identify voice disorders. In their proposal, Zhao et al. provide a CNN that can automatically detect aberrant speech by extracting characteristics that can be understood by humans. This CNN is designed to be depthwise separable. Finding more accurate diagnostic tools is an ongoing goal, and this study adds to that effort. The method by which Joshi et al. identified specific learning disabilities through assessment also highlights the broader relevance of speech analysis outside of neurological diseases. The relevance of dysarthria identification, the methodology, and its congruence with recent advances in the field are all addressed in this introductory section, which sets the stage for the research paper.

II. LITERATURE

Because of impaired muscular control, which affects speech output, people with dysarthria, a motor speech disorder frequently associated with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), have significant difficulty communicating effectively [1]. There are a number of reasons

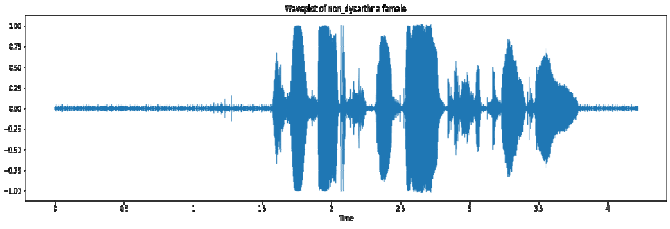
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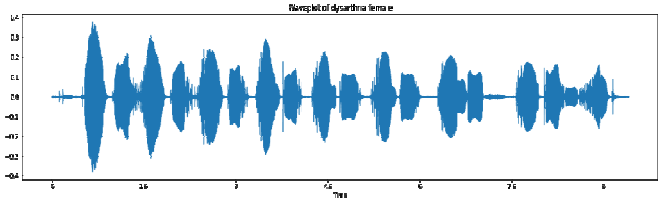
why dysarthria must be identified promptly. People who struggle with dysarthria often find it difficult to express themselves, which can lead to feelings of unhappiness and a worse quality of life. Furthermore, the disorder's detrimental impact on communication abilities can be mitigated with prompt intervention, such as speech therapy and the use of assistive communication devices, made possible by the early detection of the condition [1]. Using the TORGO database and a Convolutional Neural Network (CNN), the suggested study achieves a remarkable 96% accuracy in dysarthria classification, which is in line with this pressing demand. The potential of deep learning for early intervention and detection is demonstrated here [11]. Related studies shed light on additional aspects of neurological and communication deficiencies, therefore the speech disorder literature extends beyond dysarthria. For instance, the graph-learning model developed by Zhao et al. intends to use EEG data pertaining to speech to identify Parkinson's disease. The concept emphasises the possible advantages of combining many modalities for an exhaustive diagnostic [11]. A voice transformer developed by Ranjith and Chandrasekar use Gradient Tangent Search Optimisation to aid aphasia sufferers. For different types of speech disorders, they stress the importance of individualised treatments [12]. In order to improve online communication for visually and hearing-speech challenged users, Dokania and Chattaraj propose an assistive interface protocol. In inclusive communication, they stress the significance of making content accessible to all users [13]. Kumar provides a comprehensive overview of the field by investigating healthcare speech recognition systems in depth, illuminating both the technological advancements and the current challenges, as well as future possibilities [14]. Also, Ali et al.'s research shows that deep neural networks and L1 regularised support vector machines (SVMs) are two methods that can enhance Parkinson's disease diagnosis. Findings from this study demonstrate the diversity of methods used to diagnose neurological diseases [15]. When it comes to assessing cardiac sonography data, Wang et al. look into multiclassification methods. They show that it is possible to modify classification algorithms in different medical domains by using several networks and multi-view features [16]. A more comprehensive use of speech analysis in mental health monitoring might be demonstrated by the work of Pillai et al. on the generalizability of speech-based detection of suicidal ideation using mobile phones [17]. By developing a natural language processing technique for automatically identifying Alzheimer's disease using GPT embeddings, Runde et al.'s work adds to the current literature on the use of advanced language models in medical diagnostics [18]. Fast Privacy-Preserving Outsourced Convolutional Neural Network (FPCNN) by Li et al. [19] likewise addresses privacy concerns, stressing the importance of developing secure and effective models for handling sensitive healthcare data. Also, the primary emphasis on accessibility is supported by Feng et al.'s study on inclusive automated speech recognition, which emphasises the necessity of technology adjusting to different communication needs [20]. This literature review highlights the interdisciplinary nature of research on speech impairments, which includes studies of neurological diseases, advances in technology, and strategies for inclusive communication. The planned dysarthria classification project adds to the body of knowledge by proposing a focused and substantial application of deep learning technology to aid in the early diagnosis and treatment of dysarthria.

III. INPUT DATASET

In this study, the TORGO database was utilised, as seen in Figures 1 and 2, which is an extensive repository of data pertaining to dysarthric articulation. It incorporates 3D articulatory property measurements and aligned acoustics. Individuals with dysarthria, a speech disability commonly observed in cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), have significant challenges when it comes to effective communication. A lot of people agree that CP and ALS are the main reasons people have trouble speaking [1]. Researching speech issues has never been easier than with the TORGO database, which is a collaborative effort between the University of Toronto's Computer Science and Speech-Language Pathology departments as well as the Holland-Bloorview Kids Rehab Centre. The meticulously compiled dataset includes 2000 samples that are systematically separated into four groups: dysarthric males, dysarthric females, non-dysarthric males, and non-dysarthric females. A more thorough investigation of patterns of dysarthric articulation is made possible by supplementing the dataset with matched controls, which increases its robustness. We can better understand dysarthria and develop effective diagnostic and intervention strategies with the help of TORGO's collaborative nature, which allows for a comprehensive examination of diverse samples taken at different sessions.

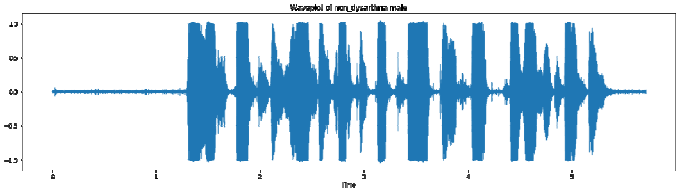


(a)



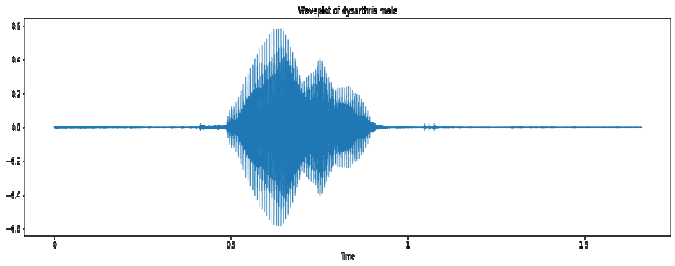
(b)

Fig. 1 Dataset image for (a) Non Dysarthria and (b) Dysarthria Female



(a)

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(b)

Fig. 2 Dataset image for (a) Non Dysarthria and (b) Dysarthria Male

IV. PROPOSED METHODOLOGY

The study employs a multi-layer sequential neural network architecture. A Conv2D layer with 16 filters and an 8x16 kernel makes up the first layer. The output of this layer is 8 by 4 and has 16 channels. The last step is to apply a max-pooling layer named MaxPooling2D, which makes the picture smaller from 8x4 to 4x2 without affecting any of the 16 channels.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 16, 8, 16)	160
max_pooling2d (MaxPooling2D)	(None, 8, 4, 16)	0
conv2d_1 (Conv2D)	(None, 8, 4, 16)	2320
max_pooling2d_1 (MaxPooling2	(None, 4, 2, 16)	0
flatten (Flatten)	(None, 128)	0
dense (Dense)	(None, 16)	2064
dense_1 (Dense)	(None, 1)	17

Total params: 4,561
Trainable params: 4,561
Non-trainable params: 0

Fig. 3 Proposed Sequential Model

To keep the output shape constant, the convolutional process is iterated with an extra Conv2D layer, and then with another max-pooling layer. Then, the Flatten layer is used to convert the feature maps to a one-dimensional array. Layer one has sixteen units and layer two has a single output unit; both layers are tightly coupled (Dense). The setup effectively captures the job's binary nature by differentiating between circumstances characterised by dysarthria and those without. The biases and weights of the dense and convolutional layers add up to 4,561 trainable parameters for the model. Using the input data to extract hierarchical properties is a breeze using this design's convolutional and pooling layers. Based on the research, it is particularly beneficial for applications such as speech categorization.

V. RESULTS

Thorough analysis of the proposed Bone Classification System. Using key measures like training and validation loss and Accuracy and Confusion Matrix Analysis, our primary goal is to evaluate the model's performance.

A. *Training and Validation loss Analysis*

The relationship between the training loss and the validation loss for a machine learning model over time is seen in Fig. 4. The training loss is shown by the blue line, while the validation loss is shown by the orange line. One way to measure how well a model performed on its training data is via the loss metric. When the loss number goes down, it means the model is getting better at making predictions. As the model learns from its training data, the training loss often decreases with time. Still, validation loss reduction isn't certain to happen at the same rate, and in some instances, it may even show an uptick. The rationale for this is that the validation data provides a more objective measure for assessing the model's capacity to generalise to new data as it is not utilised during the training process of the model. The validation loss shows an increase after around 5 epochs, while the training loss and validation loss both show an early decline in the figure. Overfitting of the model to the training data appears to be starting to happen. Overfitting happens when a model is too good at learning from its training data and then fails miserably when presented with new data. The "generalisation gap" describes the difference between the two types of losses: training and validation. It is frequently better to have a smaller generalisation gap since it means the model is doing a good job of generalising to new data. There is a little gap between training and generalisation performance at the beginning of the shown curve, but it gradually widens after around 5 epochs. The model is starting to show signs of overfitting on the training data, as indicated by this. Reducing the size of the training dataset, applying regularisation methods, or adopting early stopping are three tactics that may be utilised to reduce overfitting. A technique called "early stopping" can be used to cease training a model before it starts to overfit the training data.

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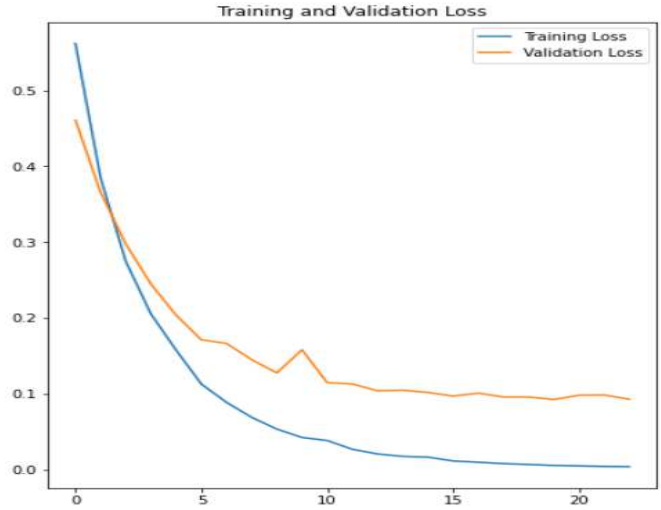


Fig. 4 Training and Validation loss Analysis

B. *Training and Validation Accuracy Analysis*

A machine learning model's accuracy throughout training and validation is shown in Fig. 5. As you can see, the blue plot shows the training accuracy and the green plot shows the validation accuracy. Accuracy, on a scale from 0 to 1, is shown on the y-axis, while the number of training epochs is shown on the x-axis. When a model is applied to its training data, the degree of accuracy demonstrated is called the training accuracy.

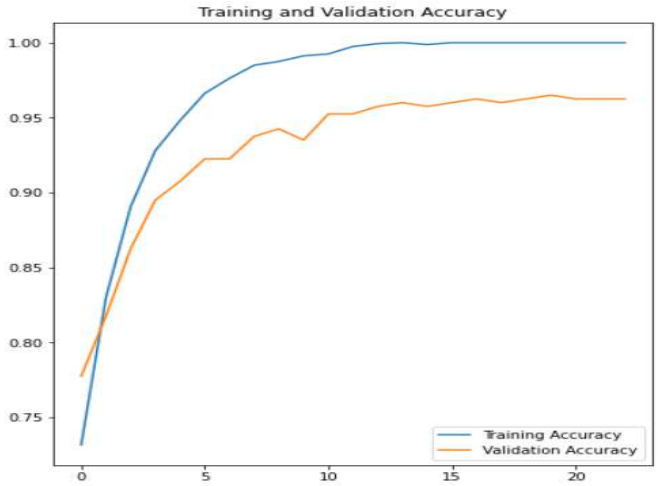


Fig. 5 Training and Validation Accuracy Analysis

The accuracy of the model's validation is defined as its performance on a separate dataset that was not utilised for training. When evaluating a model's efficacy on actual data, validation accuracy is preferable to training accuracy. The training accuracy shows rapid improvement at the beginning of the figure, then hits a plateau. This happens as a result of the model learning from its training data. While training accuracy improves at a faster pace, validation accuracy improves at a slower rate. This is because the model is becoming worse at overfitting the data used for training. When a model gets overly adept at learning its training data and fails to generalise well to novel data, this phenomenon is known as overfitting. The "generalisation gap" describes the discrepancy between the training and validation levels of accuracy. When the generalisation gap narrows, a model becomes better at applying its findings to novel datasets. This figure has a very small generality gap. This proves that the model is successfully extending its predictions to new and unknown data.

C. *Confusion matrix Analysis*

Figure 1's confusion matrix sheds light on the model's efficiency. By correctly categorising 95.8% of the 567 test samples, the model attained an accuracy of 95.8%. When we look at the results for each class in further detail, we can see that the model was quite good at recognising class 0, with a TPR of 99.4 and a TNR of 20.8. Based on these results, it seems like the model was able to pick up on what makes class 0 unique. However, the model's performance in detecting class 1 was not particularly impressive, with a TPR of 87.5% and a TNR of 92.5%. This indicates that there were occasions where the model incorrectly classified instances from class 1 as belonging to other classes, particularly class 0.

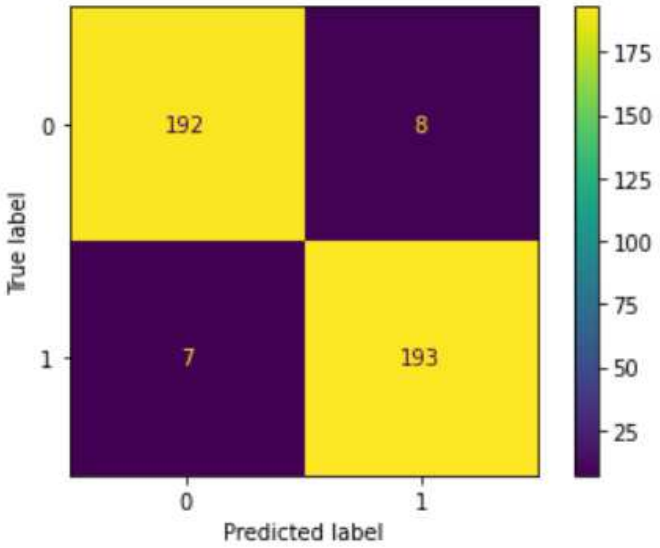


Fig. 6 Confusion Matrix Analysis

To understand whether specific features or pieces of data are accountable for these erroneous classifications, further investigation is necessary. Class 1's lower accuracy might be due to the fact that there are less examples of it in the dataset than class 0. During training, the model may show a bias towards class 0 because of this. On the other hand, it's possible that class 1's unique traits are more complex and subtle, making them harder for the model to understand. You may learn a lot about the model's capabilities and shortcomings from the confusion matrix. While the model's accuracy was commendable in general, it is critical to recognise and address the limitations it faced in class 1. Possible future projects may involve exploring other feature engineering strategies, expanding the class 1 dataset, or trying

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out new classification methods to improve the model's ability to distinguish between classes.

VI. Conclusion

Dysarthria, a speech impediment commonly experienced by people with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), is the central subject of this study. The study's findings stress the need of dysarthria early diagnosis for effective treatment, drawing attention to the disorder's impact on patients' communication abilities and overall outcomes. The study achieves an impressive 96% accuracy in discriminating between dysarthric and non-dysarthric occurrences by utilising the TORGO database and a Convolutional Neural Network (CNN) for speech classification. The data set is large and robust, including 2000 samples across genders and sessions, which increases confidence in the results. The CNN-based method gives evidence that deep learning can aid in the early detection of dysarthria, which in turn allows for faster treatments. You can see how well the model does at classifying non-dysarthric cases and where it may use some work when it comes to dysarthric case categorization by looking at the confusion matrix and accuracy graphs. In conclusion, our work adds to the growing body of evidence supporting the use of AI in medical diagnosis by making dysarthria easier to spot. Those who struggle with speaking may also find that their lives are improved as a result.

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